

The background image is a photograph of an ornate interior space, likely a courtyard or a large hall in a historical Islamic building. The walls and ceiling are covered in intricate carvings and patterns. There are several large, arched openings, some of which are filled with dark, patterned screens. The architecture features a mix of light-colored stone or plaster and darker, possibly wood or metal, elements. The overall style is highly detailed and characteristic of Islamic art.

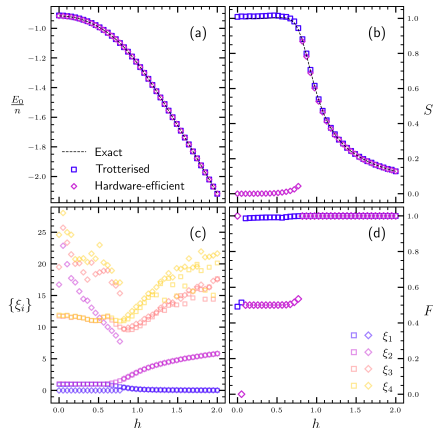
When energy isn't enough:
Certification failure in variational principle observables

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Variational algorithms assume that closeness in energy implies closeness in state. **This is not always true!**

- ▶ Low-lying energy spectrum can be quasi-degenerate even low-lying observable spectrum is not
- ▶ When spectral gap is finer than effective resolution, the correctness of an observable expectation cannot be guaranteed
- ▶ The intuition is obvious but the consequences are not formalised



If variational methods successfully minimise energy to within some effective resolution ϵ , the variational ground state $\rho \in \mathcal{H} \otimes \mathcal{H}^\dagger$ satisfies $E_0 \leq \text{Tr}(H\rho) \leq E_0 + \epsilon$. Let us define the set of computationally indistinguishable variational ground states

$$\mathcal{M}_\epsilon = \{\rho : E_0 \leq \text{Tr}(H\rho) \leq E_0 + \epsilon\}$$

The largest possible ambiguity in an observable of this state, $\mathcal{O}$, compatible with “successful” energy minimisation is given by

$$\delta_{\mathcal{O}}(\mathcal{M}_\epsilon) \equiv \sup_{\rho \in \mathcal{M}_\epsilon} |\text{Tr}(\mathcal{O}\rho) - \text{Tr}(\mathcal{O}\rho_0)| \quad (1)$$

We demonstrate through $\epsilon - \delta$ type relations that δ can be orders of magnitude larger than ϵ leading to non-linear error amplification when it is least expected

Trivially, from $F = |\langle \psi_0 | \psi \rangle|^2$, we establish the bound

$$1 - F \leq \frac{\epsilon}{\Delta}$$

and note that this can be understood in the context of populations and coherences as $1 - F = p$ is the excited population. The Fuchs–van de Graaf inequality gives

$$T \equiv \frac{1}{2} \|\rho - \rho_0\|_1 \leq \sqrt{\frac{\epsilon}{\Delta}} \quad (2)$$

and by the Holder inequality, we can write

$$\delta_{\mathcal{O}} \leq 2 \|\mathcal{O}\|_{\infty} \sqrt{\frac{\epsilon}{\Delta}} \quad (3)$$

Theorem

Let $\mathcal{O}$ be a bounded operator with certification radius (1). Then

$$\delta_{\mathcal{O}}(\mathcal{M}_{\epsilon}) \leq \epsilon \left\| H_{\text{excited}}^{-\frac{1}{2}} \mathcal{O}_{\text{excited}} H_{\text{excited}}^{-\frac{1}{2}} \right\|_{\infty} + 2\sqrt{\epsilon} \left(\sum_{n>0} \frac{|\mathcal{O}_{0n}|^2}{E_n - E_0} \right)^{\frac{1}{2}} \quad (4)$$

Proof.

See slide 7 onwards



Applying $\Delta \leq E_n - E_0$ to (4) we recover the coarse universal bound (3) from earlier:

$$\delta_{\mathcal{O}}(\mathcal{M}_{\epsilon}) \leq \mathcal{O}(\epsilon) + 2\|\mathcal{O}\|_{\infty} \sqrt{\frac{\epsilon}{\Delta}}$$

$$\delta_{\mathcal{O}}(\mathcal{M}_{\epsilon}) \leq \underbrace{\epsilon \left\| H_{\text{excited}}^{-\frac{1}{2}} \mathcal{O}_{\text{excited}} H_{\text{excited}}^{-\frac{1}{2}} \right\|_{\infty}}_{\text{population}} + \underbrace{2\sqrt{\epsilon} \left(\sum_{n>0} \frac{|\mathcal{O}_{0n}|^2}{E_n - E_0} \right)^{\frac{1}{2}}}_{\text{coherence}}$$

- ▶ In the contexts we consider, the coherence (square-root) term is the dominant contribution whenever $\sum_{n>0} \frac{|\mathcal{O}_{0n}|^2}{E_n - E_0}$ is non-zero
- ▶ The coherence term is itself dominated by low-excited state contributions

Proof of Refined Bound I

Let the ground and excited subspace projectors be

$$\rho_0 \equiv |\psi_0\rangle \langle \psi_0|, \quad \rho_{\text{excited}} \equiv \mathbb{I} - \rho_0$$

Within the excited subspace, we define an excitation Hamiltonian

$$H_{\text{excited}} = \rho_{\text{excited}}(H - E_0\mathbb{I})\rho_{\text{excited}}$$

Equivalently, we can write

$$H_{\text{excited}} = \sum_{n>0} (E_n - E_0) |\psi_n\rangle \langle \psi_n|$$

and the inverse follows as

$$H_{\text{excited}}^{-1} = \sum_{n>0} \frac{|\psi_n\rangle \langle \psi_n|}{E_n - E_0}$$

Any pure state in $\mathcal{M}_\epsilon$ satisfies the energy condition

$$\langle \psi | H | \psi \rangle \leq E_0 + \epsilon$$

and admits a decomposition as

$$|\psi\rangle = \sqrt{1-p} |\psi_0\rangle + \sqrt{p} |\chi\rangle$$

where $|\chi\rangle \in \mathcal{H}_{\text{excited}}$. It is sufficient to optimise over pure states, since $\rho \mapsto \text{Tr}(\mathcal{O}\rho)$ is affine and $\mathcal{M}_\epsilon$ is convex. The radius then looks like

$$\delta_{\mathcal{O}}(\mathcal{M}_\epsilon) = \sup_{|\psi\rangle \in \mathcal{M}_\epsilon} |\langle \psi | \mathcal{O} | \psi \rangle - \langle \psi_0 | \mathcal{O} | \psi_0 \rangle|$$

The expectation of an observable of this state follows

$$\langle \psi | \mathcal{O} | \psi \rangle = (1 - p) \langle \psi_0 | \mathcal{O} | \psi_0 \rangle + p \langle \chi | \mathcal{O} | \chi \rangle + 2\sqrt{p(1-p)} \Re \langle \psi_0 | \mathcal{O} | \chi \rangle$$

such that

$$\langle \psi | \mathcal{O} | \psi \rangle - \langle \psi_0 | \mathcal{O} | \psi_0 \rangle = p (\langle \chi | \mathcal{O} | \chi \rangle - \langle \psi_0 | \mathcal{O} | \psi_0 \rangle) + 2\sqrt{p(1-p)} \Re \langle \psi_0 | \mathcal{O} | \chi \rangle$$

The first term is a population contribution which scales as p while the second is a coherence contribution which scales as $\sqrt{p}$. We now bound the population term. Analogous to the excited Hamiltonian, we define

$$\mathcal{O}_{\text{excited}} \equiv \rho_{\text{excited}} (\mathcal{O} - \mathcal{O}_{00} \mathbb{I}) \rho_{\text{excited}}$$

Then

$$\begin{aligned}
 \langle \chi | \mathcal{O} | \chi \rangle - \langle \psi_0 | \mathcal{O} | \psi_0 \rangle &= \langle \chi | \mathcal{O}_{\text{excited}} | \chi \rangle \\
 &= \langle \chi | H_{\text{excited}}^{\frac{1}{2}} H_{\text{excited}}^{-\frac{1}{2}} \mathcal{O}_{\text{excited}} H_{\text{excited}}^{-\frac{1}{2}} H_{\text{excited}}^{\frac{1}{2}} | \chi \rangle \\
 &= \langle \chi H_{\text{excited}}^{\frac{1}{2}} | H_{\text{excited}}^{-\frac{1}{2}} \mathcal{O}_{\text{excited}} H_{\text{excited}}^{-\frac{1}{2}} | H_{\text{excited}}^{\frac{1}{2}} \chi \rangle \\
 &\leq \left\| H_{\text{excited}}^{-\frac{1}{2}} \mathcal{O}_{\text{excited}} H_{\text{excited}}^{-\frac{1}{2}} \right\|_{\infty} \langle \chi H_{\text{excited}}^{\frac{1}{2}} | H_{\text{excited}}^{\frac{1}{2}} \chi \rangle \\
 &\leq \left\| H_{\text{excited}}^{-\frac{1}{2}} \mathcal{O}_{\text{excited}} H_{\text{excited}}^{-\frac{1}{2}} \right\|_{\infty} \epsilon p^{-1}
 \end{aligned}$$

Thus

$$p(\langle \chi | \mathcal{O} | \chi \rangle - \langle \psi_0 | \mathcal{O} | \psi_0 \rangle) \leq \epsilon \left\| H_{\text{excited}}^{-\frac{1}{2}} \mathcal{O}_{\text{excited}} H_{\text{excited}}^{-\frac{1}{2}} \right\|_{\infty}$$

We now consider the coherence term, $\langle \psi_0 | \mathcal{O} | \chi \rangle$. Since $|\chi\rangle \in \mathcal{H}_{\text{excited}}$, $|\chi\rangle = \rho_{\text{excited}} |\chi\rangle$. Then

$$\begin{aligned}\langle \psi_0 | \mathcal{O} | \chi \rangle &= \langle \psi_0 | \mathcal{O} \rho_{\text{excited}} | \chi \rangle \\ &= \langle \psi_0 | \mathcal{O} \rho_{\text{excited}} \mathbb{I}_{\text{excited}} | \chi \rangle \\ &= \langle \psi_0 | \mathcal{O} \rho_{\text{excited}} H_{\text{excited}}^{-\frac{1}{2}} H_{\text{excited}}^{\frac{1}{2}} | \chi \rangle \\ &= \left(H_{\text{excited}}^{-\frac{1}{2}} \rho_{\text{excited}} \mathcal{O} | \psi_0 \rangle, H_{\text{excited}}^{\frac{1}{2}} | \chi \rangle \right)\end{aligned}$$

By the Cauchy-Schwarz inequality, we see

$$\begin{aligned} |\langle \psi_0 | \mathcal{O} | \chi \rangle| &\leq \left\| H_{\text{excited}}^{-\frac{1}{2}} \rho_{\text{excited}} \mathcal{O} | \psi_0 \rangle \right\| \left\| H_{\text{excited}}^{\frac{1}{2}} | \chi \rangle \right\| \\ &= \left\| H_{\text{excited}}^{-\frac{1}{2}} \rho_{\text{excited}} \mathcal{O} | \psi_0 \rangle \right\| (\langle \chi | H_{\text{excited}} | \chi \rangle)^{1/2} \\ &\leq \left\| H_{\text{excited}}^{-\frac{1}{2}} \rho_{\text{excited}} \mathcal{O} | \psi_0 \rangle \right\| \epsilon^{\frac{1}{2}} p^{-\frac{1}{2}} \end{aligned}$$

where the last line is obtained by expanding (5) to find that

$$p \langle \chi | H_{\text{excited}} | \chi \rangle \leq \epsilon$$

Thus

$$\left| 2\sqrt{p(1-p)} \Re \langle \psi_0 | \mathcal{O} | \chi \rangle \right| \leq 2\sqrt{\epsilon} \left\| H_{\text{excited}}^{-\frac{1}{2}} \rho_{\text{excited}} \mathcal{O} | \psi_0 \rangle \right\|$$

In an energy eigenbasis, we can expand and apply the resolution of unity to find

$$\left\| H_{\text{excited}}^{-\frac{1}{2}} \rho_{\text{excited}} \mathcal{O} | \psi_0 \rangle \right\|^2 = \sum_{n>0} \frac{|\mathcal{O}_{0n}|^2}{E_n - E_0}$$

Combining the population and coherence contributions yields the radius bound

$$\delta_{\mathcal{O}}(\mathcal{M}_{\epsilon}) \leq \epsilon \left\| H_{\text{excited}}^{-\frac{1}{2}} \mathcal{O}_{\text{excited}} H_{\text{excited}}^{-\frac{1}{2}} \right\|_{\infty} + 2\sqrt{\epsilon} \left(\sum_{n>0} \frac{|\mathcal{O}_{0n}|^2}{E_n - E_0} \right)^{\frac{1}{2}}$$

- ▶ By Ehrenfest's theorem, $\sum_{n>0} \frac{|\mathcal{O}_{0n}|^2}{E_n - E_0} = \hbar^2 \sum_{n>0} \frac{|\dot{\mathcal{O}}_{0n}|^2}{(E_n - E_0)^3}$ such that the coherence channel is suppressed for conserved quantities
- ▶ Approximate conservation suppresses the bound by powers of the gap
- ▶ If S is a symmetry such that $[H, S] = 0$, then the relevant spectral gap is restricted to the ansatz symmetry sector $\Delta_{s_0} \equiv \min_{s_n=s_0, n>0} (E_n - E_0) > \Delta$
- ▶ If $\rho_{s_0} \mathcal{O} |\psi_0\rangle = 0$, the coherence channel vanishes
- ▶ Under weak-symmetry breaking, $\ell(\psi) = \|(1 - \rho_{s_0}) |\psi\rangle\|^2$, a coherence contribution of $2\|\mathcal{O}\|_\infty \sqrt{\ell}$ is reintroduced

Numerous implementations of the variational principle exist, each with its own effective resolution, ϵ_{eff} . Loosely, certification fails once $\Delta \lesssim \epsilon_{\text{eff}}$

- ▶ For an algorithm with fixed $\epsilon_{\text{eff}} = \epsilon_{\text{chem}} \approx 1.6\text{m}E_h$ we lower bound the the certifiable regime by a spectral gap $\Delta_{\text{cert}} \approx 1.6\text{m}E_h$
- ▶ The theorem demonstrates that a simulation may produce large observable errors even when energy is within chemical accuracy

We consider three such implementations

Algorithm	State-of-art $\epsilon_{\text{eff}} \backslash \Delta_{\text{cert}}$	Ref.
VQE	$\approx 1.6\text{m}E_h$	[1]
DMRG	$\approx 10^{-3}\text{m}E_h$	[2]
SqDRIFT	$\approx 20\text{m}E_h$	[3]

These numbers may improve with hardware but the algorithms have fundamental limits. We investigate how these limits vary with system size

- ▶ Near criticality, the spectral gap generally closes as $\Delta(L) \sim L^{-\gamma}$, where γ is the system critical exponent
- ▶ The ratio $r = \lim_{L \rightarrow \infty} \frac{\epsilon_{\text{eff}}(L)}{L^{-\gamma}}$ predicts when observable ambiguity will dominate, allowing us to make concrete statements about certifiability in the asymptotic limit

Note: Criticality leading to quasi-degeneracy is generally understood in condensed matter but less so in chemistry. These quasi-degeneracies are also expected in a multireference molecule, where the number of electronic configurations grows combinatorially with size such that we should expect a closing spectral gap. Polymers may serve as an ideal molecular example here

$$\epsilon_{\text{VQE}} = \epsilon_{\text{ansatz}} + \epsilon_{\text{opt}} + \epsilon_{\text{stat}} + \epsilon_{\text{noise}} + \epsilon_{\text{Ham}}$$

- ▶ In an ideal context, some of these errors could be eliminated entirely ($\epsilon_{\text{ansatz}} = \epsilon_{\text{opt}} = \epsilon_{\text{Ham}} = 0$)
- ▶ What is leftover is fundamental to the algorithm, $\epsilon_{\text{VQE}} \sim \frac{\Lambda_H(L)}{\sqrt{N_{\text{shot}}}} + \eta G(L)$ [4]
- ▶ Λ_H generically scales as $\mathcal{O}(L^4)$ but is lower bounded by $\mathcal{O}(L)$ [5] whilst $G(L)$ is shown to scale as $\mathcal{O}(L^4)$ under UCCSD [6]
- ▶ Regardless, these are increasing such that effective resolution grows with system size and certifiability is lost in the asymptotic limit ($r \rightarrow \infty$) for all $\gamma \geq 0$

As system size increases, asymptotic certification remains possible if energy resolution decreases faster than the spectral gap closes

$$\epsilon_{\text{DMRG}} = \epsilon_{\chi} + \epsilon_{\text{sweep}} + \epsilon_{\text{Ham}}$$

- ▶ Ideally, ϵ_{sweep} and ϵ_{Ham} can be made negligible
- ▶ Let $w_{\chi}(L) = 1 - \sum_{j=1}^{\chi} \lambda_j^2$ denote the discarded Schmidt weight at χ [7]
- ▶ Heuristically, $\epsilon_{\chi}(L) \sim L w_{\chi}(L)^{\beta}$ with support for $\beta = 1$ [8, 9], $\beta = \frac{1}{2}$ [9], etc. depending on context
- ▶ Certification ($r = 0$) therefore requires $w_{\chi}(L) \lesssim \mathcal{O}(L^{-(1+\gamma)/\beta})$

in prog.

$$\mu_i = - \sum_a r_{a,i} + \sum_A Z_A R_{A,i}$$

- ▶ We observe generic certification within $\delta_\mu \lesssim \mathcal{O}(|\mu_{0n}|\sqrt{\frac{\epsilon}{\Delta}})$
- ▶ If symmetry (selection-rules) enforce $\mu_{0n} = 0$ for low-lying states, then the leading coherence channel vanishes, improving scaling to $\mathcal{O}(\frac{\epsilon}{\Delta})$
- ▶ During charge-transfer, the matrix elements can scale as $\mu \sim \mathcal{O}(L)$, significantly worsening certification in the asymptotic limit regardless of the algorithm
- ▶ LiF sees an ionic-covalent avoided crossing at $r_c = 13.7a_0$ with $\Delta = 1.27\text{m}E_h$ [10]. Given that $|\mu_{01}| \sim \frac{1}{2}er_c \approx 17\text{D}$ [11], if $\epsilon = \epsilon_{\text{chem}} \approx 1.6\text{m}E_h$ then $\delta_\mu \lesssim 39\text{D}$, where the true dipole is $\approx 17\text{D}$ (bound spans the entire physical range)

$$\alpha_{ij} = 2 \sum_{n \neq 0} \frac{\langle \psi_0 | \mu_i | \psi_n \rangle \langle \psi_n | \mu_j | \psi_0 \rangle}{E_n - E_0}$$

- ▶ This is a second-order response function such that error scaling is more severe, $\delta_\alpha \lesssim \mathcal{O}\left(\frac{|\mu_{0n}|^2 \sqrt{\epsilon}}{\Delta^{3/2}}\right)$
- ▶ More generally, the k -order response contains k transition elements and $k - 1$ denominators, so near-degenerate amplification follows $R^{(k)} \lesssim \mathcal{O}\left(\frac{|O_{0n}|^k \sqrt{\epsilon}}{\Delta^{k-1}}\right)$
- ▶ Symmetry can still protect the response (i.e. $\mu_{0n} = 0$) but in the other direction, strong scalings are worsened

$$S = -\text{Tr} \left(\rho_{\frac{1}{2}}^{\text{red}} \log \rho_{\frac{1}{2}}^{\text{red}} \right)$$

- ▶ From (2) and the Fannes-Audenaert inequality, we directly bound the certifiable radius as $\delta_S \lesssim \mathcal{O} \left(\log d \sqrt{\frac{\epsilon}{\Delta}} \right)$, where $d = \dim \mathcal{H}_A$
- ▶ This is not “technically” an observable, it is a non-linear functional of the reduced state, but the logic still carries forward
- ▶ The inequality saturates at $\delta_S = \frac{L}{2}$ for maximally distant reduced density matrices, though in practice this is never achieved as the Hilbert space is restricted by physical constraints

- ▶ Minimisation of energy does not guarantee correct observable expectation values
- ▶ Additional conditions ($\epsilon_{\text{eff}} < \Delta$ and well-behaved $\mathcal{O}_{0n}$) are required for that
- ▶ This is really a problem near criticality and in large multi-reference systems
- ▶ To reiterate, our claim is not that these algorithms must fail but that if they did fail we would not necessarily know such that their results are not certifiable
- ▶ Most useful quantum chemistry predictions come directly from the potential energy surface so this is not the end of the world

Going forward

- ▶ Excited states and general correlation functions!
- ▶ Formulation in terms of static susceptibility
- ▶ Applications to specific systems rather than general heuristic?
- ▶ When is failure guaranteed? (outside the scope of this framework)
- ▶ Proof that asymptotic multireference systems have an exponentially closing gap

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